

tacit knowledge, the idiosyncratic tips and tricks that accumulate with experience.⁵ As young workers supply relatively more codified knowledge than tacit knowledge, they may face greater task replacement from AI in exposed occupations, leading to greater employment reallocation (Acemoglu and Autor, 2011). In contrast older workers with accumulated tacit knowledge may face less task replacement. These benefits of tacit knowledge may accrue less to non-college workers in occupations with low returns to experience. Furthermore, more experienced workers may be more skilled in other ways, making them less vulnerable to substitution by AI tools (Ide, 2025). An important direction for research is to further model and test these predictions.

2 Related Literature

This paper engages with a large public debate in academia, public policy, business, and media on the employment effects of artificial intelligence. Much of this discourse centers on whether AI is displacing workers in exposed professions such as software engineers.⁶ Some work has noted that the unemployment rate for college graduates has risen above the rate for non-graduates, suggesting this as evidence of employment disruptions from AI (Thompson, 2025). Others have noted that these trends long preceded the spread of AI and have noted that publicly available data such as the Current Population Survey (CPS) show mixed evidence on employment changes in AI-exposed occupations (Lim et al., 2025; The Economist, 2025; Smith, 2025; Eckhardt and Goldschlag, 2025; Frick, 2025).⁷ These debates remain unsettled and in search of high quality data on labor market changes among exposed groups. Our paper provides large-scale data to measure employment changes with a high degree of granularity and precision, finding that young workers in AI-exposed occupations have indeed experienced employment declines.

⁵Ironically, one of the practical skills more likely to be learned on the job than in university computer science classes may be how to use AI software development.

⁶Some recent media on this topic includes Horowitz (2025); Ettenheim (2025); Raman (2025); Roose (2025); Peck (2025); Hoover (2025); Milmo and Almeida (2025); Wu (2025); Raval (2025). A number of technology executives have also warned of potential job loss from AI (Allen, 2025; Sherman, 2025; Bacon, 2025) or laid off workers with the aim of increasing AI investments (Jamali, 2025).

⁷Reports from industry have also shown mixed findings. Job posting platform TrueUp suggests a recent increase in postings in the tech sector (Lenny Rachitsky, 2025). On the other hand, Revelio Labs finds a decline in job postings, with the decrease steeper for entry-level workers (Simon, 2025). Indeed job posting data suggest declines in postings for new graduates but find these declines for less AI-exposed occupations as well (Lim et al., 2025). Chandar (2025b) notes that the correlation between job postings and employment has been weak over recent years. SignalFire finds steep declines in new graduate hires in the tech sector compared to pre-Pandemic levels (Doshay and Bantock, 2025), consistent with the findings in this paper. Data from Gusto also suggests a decline in new graduate hiring (Bowen, 2025).

In academia, there is a growing body of research seeking to measure the employment effects of AI. The advent of this literature included a series of influential papers that established methodologies for estimating which occupations and tasks were susceptible to automation (Frey and Osborne, 2017; Brynjolfsson and Mitchell, 2017; Brynjolfsson et al., 2018; Felten et al., 2018, 2019; Webb, 2019; Felten et al., 2021). More recently, work such as Eloundou et al. (2024); Felten et al. (2023); Gmyrek et al. (2023); Handa et al. (2025), and Tomlinson et al. (2025) adapted this approach for Generative AI, forming the basis for exposure metrics used in this analysis. While these studies identify potential disruption, ours connect these exposure measures to actual employment changes. We find that these measures of exposure do indeed predict substantial employment changes for young workers in the period after the spread of generative AI.

Our work complements and broadens insights from studies that find significant effects in more specific settings, such as on online freelance platforms (Hui et al., 2023; Demirci et al., 2025) or within individual firms (Brynjolfsson et al., 2025; Dillon et al., 2025).⁸ We measure labor market changes across occupations spanning the US economy.

In this sense our work complements a small but growing list of papers that use economy-wide data to measure AI’s impact. Recent findings have been varied. Humlum and Vestergaard (2025) use Danish administrative data to conclude there were minimal effects on earnings or hours worked, while Jiang et al. (2025) find AI exposure is correlated with longer work hours in the U.S.⁹ Hampole et al. (2025) use job postings and LinkedIn profile records from Revelio labs from 2011 to 2023 to find limited employment impacts overall, with growing overall labor demand at firms offsetting relative declines in demand for exposed occupations. Chandar (2025b) uses data from the CPS to compare employment changes in more and less AI-exposed professions, finding little differential trend overall but noting the difficulty of measuring changes for young workers because of the limited effective sample size. Dominski and Lee (2025) similarly use CPS data with alternative exposure measures and find declines in employment in AI-exposed occupations, though data limitations in the CPS limit the capacity for statistical inference. Johnston and Makridis (2025) find employment increases in state-industry pairs more exposed to AI using data from the

⁸See also Noy and Zhang (2023); Peng et al. (2023); Dell’Acqua et al. (2023).

⁹See also Acemoglu et al. (2022); Bonney et al. (2024); Bick et al. (2024); Hartley et al. (2025); Frank et al. (2025); Chen et al. (2025).

Quarterly Census of Employment and Wages (QCEW).¹⁰ These prior papers use data that lack either sufficient granularity or immediacy to reliably study employment changes by AI exposure and age (O’Brien, 2025).¹¹ In contrast, this paper uses large-scale, close to real-time data to take a step towards resolving the ongoing debate on the employment effects of AI on young workers.

3 Data Description

3.1 Payroll Data

This study uses data from ADP, the largest payroll processing firm in America. The company provides payroll services for firms employing over 25 million workers in the US. We use this information to track employment changes for workers in occupations measured as more or less exposed to artificial intelligence.

We make several sample restrictions for our main analysis sample. We include only workers employed by firms that use ADP’s payroll product to maintain worker earnings records. We also exclude employees classified by firms as part-time from the analysis and subset to people between the age of 18 and 70.¹²

The set of firms using payroll services changes over time as companies join or leave ADP’s platform. We maintain a consistent set of firms across our main sample period by keeping only companies that have employee earnings records for each month from January 2021 through July 2025.

In addition, ADP observes job titles for about 70% of workers in its system. We exclude workers who do not have a recorded job title. There are over 7,000 standardized job titles, examples of which include “Search engineer optimization specialist,” “Enterprise content management manager,” and “Plant documentation control specialist.” The company’s internal research team maps each of

¹⁰Johnston and Makridis (2025) measure state-industry exposure by taking an average of Eloundou et al. (2024)’s occupational exposure weighted by state-industry employment. Industry-level labor market changes may be distinct from the occupation-level changes studied in this paper if firms make capital investments or become more productive in ways that increases overall labor demand (Hampole et al., 2025).

¹¹As a comparison, the CPS surveyed between 44,000 and 51,000 employed individuals in total across all age groups in each month since 2021. Between 10,000 and 12,000 of these observations were in the outgoing rotation group and included earnings records. The data in our main analysis sample includes between 250,000 and 350,000 employed individuals in each month *just between the ages of 22 and 25*, all with earnings records.

¹²While we observe the year of birth for each worker, for privacy reasons we do not observe the exact date of birth. We impute month of birth from the distribution of birth months in the United States using data from the Center for Disease Control and Prevention.

these job titles to a 2010 Standard Occupational Classification (SOC) code, additionally using information such as the job description, industry, location, and other relevant data. We use these estimated SOC codes to merge our data to occupational AI exposure measures described below in Section 3.2.

After these restrictions we have records on between 3.5 and 5 million workers each month for our main analysis sample, though we consider robustness to alternative analyses such as allowing for firms to enter and leave the sample.

While the ADP data include millions of workers in each month, the distribution of firms using ADP services does not exactly match the distribution of firms across the broader US economy. Further details on differences in firm composition can be found in [Cajner et al. \(2018\)](#) and [ADP Research \(2025\)](#).¹³

3.2 Occupational AI Exposure

We use two different approaches for measuring occupational exposure to AI. The first uses exposure measures from [Eloundou et al. \(2024\)](#). [Eloundou et al. \(2024\)](#) estimate AI exposure by O*NET task using ChatGPT validated with human labeling. They then construct occupational exposure measures by aggregating the task data to the 2018 SOC code level. We focus on the GPT-4 based β exposure measures from their paper.

The second primary approach we take uses data on generative AI usage from the Anthropic Economic Index ([Handa et al., 2025](#)). This index reports the estimated share of queries pertaining to each O*NET task based on a sample of several million conversations with Claude, Anthropic’s generative AI model. It then aggregates the data to the occupational level based on these task shares. One feature of the Anthropic Economic Index is that for each task it also reports estimates of the share of queries pertaining to that task that are “automotive,” “augmentative,” or none of the above. We use this information as an estimate of whether usage of AI for an occupation is primarily complementary or substitutable with labor.¹⁴

¹³[Cajner et al. \(2018\)](#) find a somewhat higher share of manufacturing and services firms compared to the Quarterly Census of Employment and Wages (QCEW) using data from March 2016. They also find that ADP somewhat overrepresents firms in the Northeast. In addition, firms using ADP tend to grow faster on average than the typical firm in the US economy.

¹⁴Specifically, [Handa et al. \(2025\)](#) first use Claude to classify conversations into six categories: Directive, meaning complete task delegation with minimal interaction; Feedback Loop, meaning task completion guided by environmental feedback such as when repeatedly relaying coding errors to the model; Task Iteration, meaning a collaborative